Predictive Modeling of *Salmonella enteritidis* Growth Under Sinusoidal Temperature

Modified Gompertz Model & OLS Parameter Estimation

Boyuan Zhao, Wenyu Xu, Yizhou Li

College of Biosystems Engineering and Food Science
Zhejiang University

April 14, 2026

目录

1 Introduction & Background

2 Model Description

3 Data Overview

4 Forward Problem

5 Parameter Selection

6 OLS Results

7 Model Diagnostics

8 Optimal Experimental Design

9 Bootstrapping

10 Conclusions

11 References

Introduction & Background

- *Salmonella enteritidis* is a major foodborne pathogen in egg products
- Temperature fluctuations during storage/transport affect microbial growth
- Reference: Gumudavelli et al. (2007), *J. Food Sci.* 72:M254–62 [1]

Objective

Use a modified Gompertz model to predict *Salmonella* growth under sinusoidal heating, estimate model parameters via OLS, and evaluate model adequacy.

Model Description

Primary Model (differential form)

$$\frac{d\log N}{dt} = \mu \cdot K \cdot C \cdot \exp(-K), \quad K = \exp\big(-\mu(t-M)\big) \tag{1}$$

Secondary Model (growth rate vs. temperature)

$$\mu = a(T - T_{\min})^2\big(1 - \exp[b(T - T_{\max})]\big) \tag{2}$$

- **Dependent variable**: $\log_{10} N(t)$ —log microbial concentration (log cfu/mL)
- **Independent variable**: t —time (hr)

Model Description —7 Candidate Parameters

Parameter	Symbol	Description	Units	Initial Guess
A	A	Initial log microbial number	$\log \mathrm{cfu/mL}$	$\log_{10}(400) \approx 2.60$
C	C	Upper–lower asymptote difference	$\log \mathrm{cfu/mL}$	11
M	M	Inflection point time	hr	7.5
a	a	Secondary model coefficient	$^\circ\mathrm{C}^{-2}\,\mathrm{hr}^{-1}$	0.000338
b	b	Secondary model coefficient	$^\circ\mathrm{C}^{-1}$	0.275
$T_{\min}$	$T_{\min}$	Minimum growth temperature	$^\circ\mathrm{C}$	6
$T_{\max}$	$T_{\max}$	Maximum growth temperature	$^\circ\mathrm{C}$	46.3

Data Overview

- Growth data: $\log_{10}(\text{CFU/mL})$, $n = 20$
- Temperature: sinusoidal $T(t)$ every minute
- Two independent time vectors

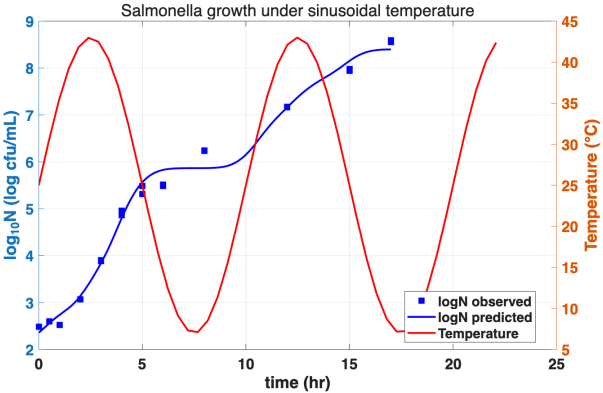

Fig 1. Dual-axis: $\log_{10} N$ (observed & predicted) and T vs. time

Forward Problem —Initial Guess

- All 7 guesses from literature
- ODE solved with `ode45`
- $T(t)$ interpolated via `interp1`

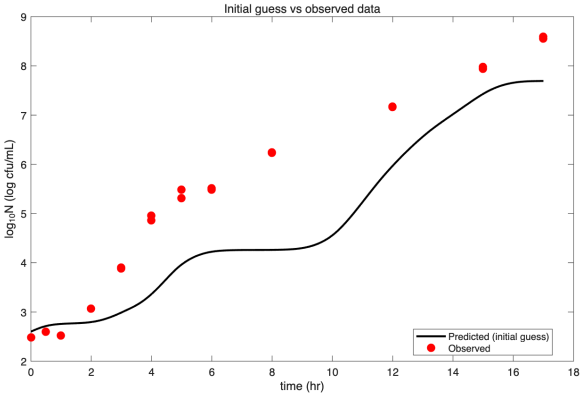

Fig 2. Initial guess vs. observed data

SSC Analysis —All 7 Parameters (Step 1)

Max|SSC|:

Param	Max|SSC|
A	2.61
C	5.11
M	5.45
a	2.20
b	**0.88**
$T_{\min}$	**0.83**
$T_{\max}$	9.02

Decision

$T_{\min}$ (0.83) and b (0.88): lowest SSC $\Rightarrow$ **fix**: $T_{\min} = 6\,^{\circ}\mathrm{C}$, $b = 0.275$.

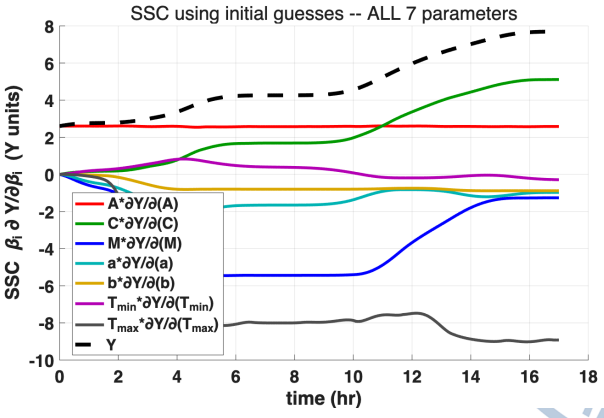

Fig 3. SSC for all 7 parameters

Parameter Selection Summary

Parameter	Max|SSC|	Identifiable?	Action	Value Used
A	2.61	✓Yes	**Estimate**	OLS result
C	5.11	✓Yes	**Estimate**	OLS result
M	5.45	✓Yes	**Estimate**	OLS result
a	2.20	✓Yes	**Estimate**	OLS result
b	0.88	× Low SSC	Fix (literature)	$0.275\,^\circ\mathrm{C}^{-1}$
$T_{\min}$	0.83	× Lowest SSC	Fix (literature)	$6\,^\circ\mathrm{C}$
$T_{\max}$	9.02	✓Yes	**Estimate**	OLS result

- **Direct elimination**: $7 \to 5$ —fix $T_{\min}$, b (low SSC)
- **Final estimable parameters**: $[A, C, M, a, T_{\max}]$

OLS Inverse Problem —Final Results ($[A, C, M, a, T_{\max}]$)

Param	Estimate	σ	Rel Err	95% CI
A	2.358	0.146	6.2%	[2.046, 2.670]
C	11.062	2.013	18.2%	[6.771, 15.354]
M	5.541	0.528	9.5%	[4.414, 6.667]
a	3.0e-4	1.1e-4	38.3%	[1.0e-4, 6.0e-4]
$T_{\max}$	45.72	9.15	20.0%	[26.23, 65.22]

- Fixed: $T_{\min} = 6\,^{\circ}\mathrm{C}$, $b = 0.275$ (literature)
- cond$(\mathbf{J}) = 3.14 \times 10^5$ RMSE $= 0.232$ relRMSE $= 3.84\%$ $R^2 = 0.989$
- Correlations: $R(C, T_{\max}) = 0.98$, $R(C, a) = -0.97$, $R(a, T_{\max}) = -0.94$, $R(C, M) = 0.92$ — all 95% CI exclude zero

Fitted Curve with Asymptotic CB and PB

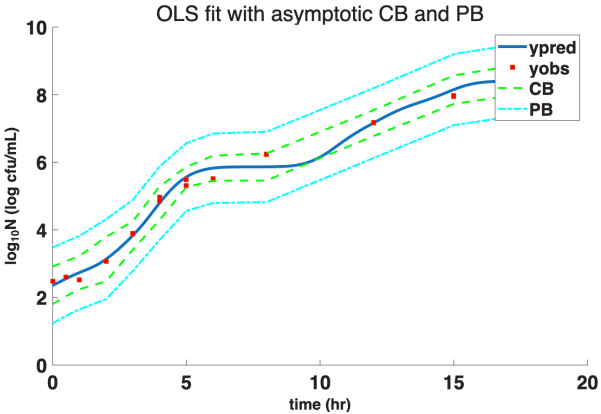

Fig 6. OLS fit with 95% simultaneous confidence bands (CB) and prediction bands (PB)

Residual Analysis

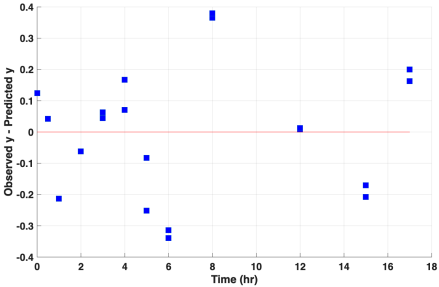

Fig 7. Residual scatter

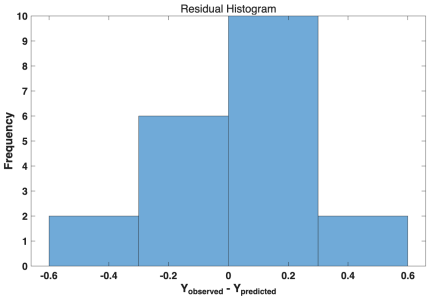

Fig 8. Residual histogram

5 Assumptions: ✓ Zero mean ($\bar{e} \approx 0.000$) △ Constant variance △ Uncorrelated (runs $= 6 < 10.5$)
✓ Normality ✓ Correct model

SSC Convergence —Before & After Estimation

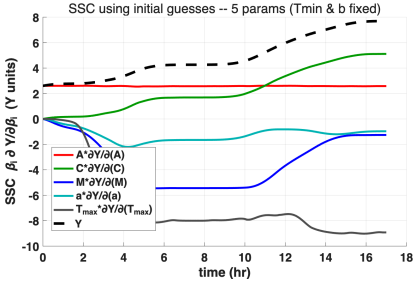

Fig 9a. SSC (initial guess)

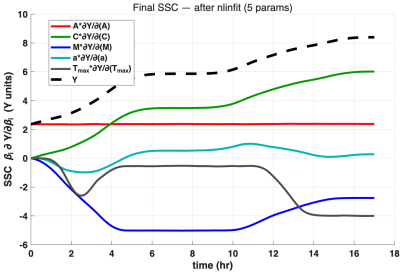

Fig 9b. SSC (after OLS)

Double-check (per inv_soln5)

A, C, M: stable. a and $T_{\max}$ reduced ∼50% (initial guess far from estimate), but all Max|SSC| $\geq 1 \Rightarrow$ identifiability **confirmed**.

Optimal Experimental Design

- Beck & Arnold (1977): $C_{ij} = \frac{1}{n}\sum X_i' X_j', \quad \Delta = \det(C)$
- Δ peaks at $n \approx 9 \Rightarrow$ optimal design
- C_{ii}: C, M dominate; a contributes least information

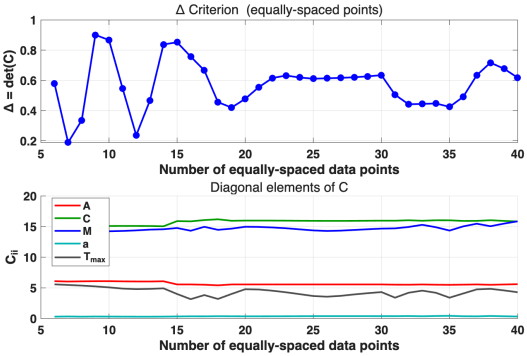

Fig 10. Δ criterion and C_{ii} vs. data points

Bootstrapping of Residuals

600 bootstrap iterations (residual resampling), 522/600 valid.

Param	Bootstrap 95% CI	Asymptotic 95% CI
A	[2.093, 2.606]	[2.046, 2.670]
C	[10.27, 12.94]	[6.771, 15.354]
M	[4.652, 5.836]	[4.414, 6.667]
a	[2.5e-4, 4.4e-4]	[1.0e-4, 6.0e-4]
$T_{\max}$	[43.26, 152.3]	[26.23, 65.22]

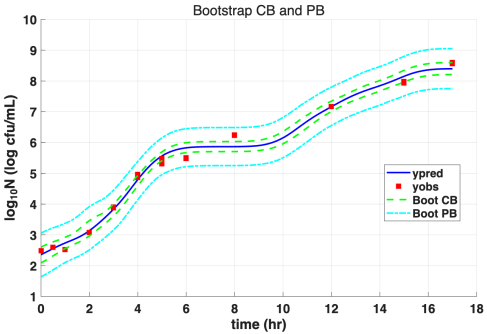

Fig 11. Bootstrap CB and PB

Asymptotic vs Bootstrap Comparison

Metric	Asymptotic	Bootstrap
CB width (avg)	0.867	**0.365**
PB width (avg)	2.119	**1.274**

- Bootstrap CB is $\mathbf{2.4}\times$ narrower; Bootstrap PB is $\mathbf{1.7}\times$ narrower
- Asymptotic method overestimates uncertainty ($\text{cond}(\mathbf{J}) = 3.14 \times 10^5$ inflates covariance)
- Bootstrap provides more realistic uncertainty quantification

Conclusions

1. Modified Gompertz model successfully predicts *Salmonella* growth under sinusoidal temperature
2. **Parameter selection**: SSC analysis $\Rightarrow$ fix $T_{\min}$, b (low SSC); estimate $[A, C, M, \alpha, T_{\max}]$ $(7 \to 5)$. $R(C, T_{\max}) = 0.980$ but all 95% CI exclude zero
3. OLS fit: cond$(\mathbf{J}) = 3.14 \times 10^5$, relRMSE $= 3.84\%$, $R^2 = 0.989$; residual autocorrelation (runs $= 6 < 10.5$)
4. Bootstrap CI narrower ($2.4\times$ for CB, $1.7\times$ for PB) $\Rightarrow$ asymptotic overestimates due to high parameter correlation

References I

[1] Lavanya Gumudavelli, Prem Rishi Velugoti, Ramaswamy C Anantheswaran, John D Floros, and Stephen J Knabel.
Dynamic predictive model for growth of *Salmonella Enteritidis* in egg yolk.
Journal of Food Science, 72(7):M254–M262, 2007.